

Data Management (A.A. 2025/26) – exam A of 4/06/2026

Problem 1 Let x and y be two elements of the database such that $rts(x) = rts(y) = wts(x) = wts(y) = wts-c(x) = wts-c(y) = 0$ and $cb(x) = cb(y) = true$. Suppose that the system clock is 0, and that the system uses the clock values as the timestamps to be assigned to the various transactions (at their first action). Illustrate the actions of the timestamp-based scheduler when receiving the following complete schedule:

$$r_1(x) \ r_2(x) \ r_3(y) \ w_2(y) \ c_2 \ w_3(x) \ c_1 \ c_3$$

and tell whether the resulting sequence of actions (obtained by ignoring aborted transactions) is a rigorous schedule.

Problem 2 Consider the following schedule S :

$$r_1(x) \ r_2(y) \ w_1(x) \ r_1(y) \ w_2(x) \ r_2(x) \ w_3(x) \ r_4(y) \ w_4(y) \ r_3(y)$$

- 2.1 Determine whether S is conflict-serializable or not, explaining your answer in detail.
- 2.2 Determine whether S is a 2PL schedule with exclusive locks. If it is not, determine whether adding shared locks would make it a 2PL schedule. Justify your answers in detail.
- 2.3 Determine whether S a strict schedule. Justify your answers in detail.

Problem 3 Let $R(A, B)$ and $S(A, B)$ be two bags with 5.000 and 15.000 pages, respectively. Assume that exactly 3 buffer frames are available. For each of the following situations:

- 3.1 R and S are not sorted;
- 3.2 R and S are sorted on all attributes;
- 3.3 R and S are sorted on attribute A ;
- 3.4 R and S are sorted on attribute A , and no two tuples of R having the same value of attribute A are stored in different pages. The same property holds for relation S .

illustrate which algorithm you would choose for computing the set intersection $R \cap S$, and tell which is the cost of the algorithm in terms of number of page accesses.

Problem 4 Let $R(A, B, C, D)$ be a relation stored in a heap with 48.000 pages. Assume that all attribute values have the same size and that 160 buffer frames are available. Consider the following query

`select distinct B, D from R`

(note that the result is without duplicates). Describe the most efficient algorithm that can be used to compute the query. Justify your choice and compute its cost in terms of number of page accesses.

Problem 5 (only for students who opted for **option 1**, i.e., who do **not** do the project)

- 5.1 Write an RDFS dataset G that expresses the following information. Use the RDF/RDFS vocabulary where appropriate; otherwise, use any URI you prefer (assuming an empty prefix):
 - Some journalist has announced that the Mona Lisa was bought by someone anonymous for 20 billion dollars.
 - The Mona Lisa was painted by Leonardo Da Vinci.
 - Leonardo Da Vinci is also the name of an airport.
 - Everything that is created is a work of art.
 - Everyone who creates is an artist.
 - For every x, y , if x is painted by y , then x is created by y .
 - Artist and journalist are humanities profession.
 - Every humanities profession is a profession.
- 5.2 Write the SPARQL query Q that retrieves all works of art.
- 5.3 Provide the answers to Q when it is (a) evaluated over the dataset G interpreted as a graph database, and (b) computed over G interpreted as an RDFS Knowledge Graph. Justify your answers.